

Study Guide



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

**Faculty of Engineering,
Built Environment and
Information Technology**

Fakulteit Ingenieurswese, Bou-omgewing en
Inligtingtegnologie / Lefapha la Boetšenere,
Tikologo ya Kago le Theknolotši ya Tshedimošo

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Department of Electrical, Electronic and Computer Engineering

Electricity and Electronics

EBN111

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1 Module calendar: important dates and overview

The module calendar provides detailed information to students on how to prepare for each contact session. Preparing before class is key for actively participating in class discussions and activities. The goal is to have everyone ready to engage when class starts.

EBN111 LECTURE SCHEDULE 2025

XY – Prof. Xianming Ye; ST – Sakhile Twala; AR – Archie Rohde; AL – Assistant Lecturers

*Venues: **Mon** - AE du Toit Auditorium; **Tue** - AE du Toit Auditorium; **Thu**: Roos hall; **Fri**: AE du Toit Auditorium*

Week	Day	Date	Venue	Instructor	Event	Lecture contents and study tasks
1	Monday 08:30 – 09:20	10/02/2025	AE du Toit Auditorium	ST	Lecture 1	Introduction and Study Guide Study Ch 1.1-1.4 and do textbook Examples: 1.1, 1.2, 1.3 Units, Charge, Current, Voltage Do textbook Practice Problems: 1.1, 1.2, 1.3 Do AMS Assignment #1 (Deadline: 17 February 2025 at 11:59pm)
	Tuesday 13:30 – 14:20	11/02/2025	AE du Toit Auditorium	ST	Lecture 2	Study Ch 1.5-1.6 and do textbook Examples: 1.4, 1.5, 1.6, 1.7 Power, Energy and Circuit Elements Do textbook Practice Problems: 1.4, 1.5, 1.6, 1.7
	Thursday 11:30 – 12:20	13/02/2025	Roos hall	ST	Lecture 3	Study Ch 2.2-2.3 and do textbook Examples: 2.1, 2.2, 2.3, 2.4 Ohm's Law, Nodes & Branches & Loops Do textbook Practice Problems: 2.1, 2.2, 2.3, 2.4
	Friday 07:30 – 08:20	14/02/2025	AE du Toit Auditorium	ST & AL	Tut class #1	Pre-practical Lecture 1 & AMS tutorial Do chapter problems: 1.4, 1.6, 1.8, 1.11, 1.12, 1.16, 1.18, 1.20, 1.24, 1.28, 2.6, 2.7 Do Pre-practical 1 Assignment and submit on AMS (Deadline: 21 February 2025 at 11:59pm)

2	Monday 08:30 – 09:20	17/02/2025	AE du Toit Auditorium	ST	Lecture 4	Study Ch 2.4 and do textbook Examples: 2.5, 2.6, 2.7, 2.8 <i>Kirchhoff's Law</i> Do textbook Practice Problems: 2.5, 2.6, 2.7, 2.8 Do AMS Assignment #2 (Deadline: 24 February 2025 at 11:59pm)
	Tuesday 13:30 – 14:20	18/02/2025	AE du Toit Auditorium	ST	Lecture 5	Study Ch 2.5-2.6 and do textbook Examples: 2.9, 2.10, 2.11, 2.12, 2.13 <i>Series and Voltage division, Parallel and Current division</i> Do textbook Practice Problems: 2.9, 2.10, 2.11, 2.12, 2.13
	Thursday 11:30 – 12:20	20/02/2025	Roos hall	ST	Lecture 6	Study Ch 2.7 and do textbook Examples: 2.14, 2.15 <i>Wye-Delta transformations</i> Do textbook Practice Problems: 2.14, 2.15
	Friday 07:30 – 08:20	21/02/2025	AE du Toit Auditorium	ST, AR & AL	Tut class #2	Collect component packs Do chapter problems: 2.15, 2.22, 2.23, 2.25, 2.32, 2.45b, 2.51b, 2.53b

Week	Day	Date	Venue	Instructor	Event	Lecture contents and study tasks
3 Practical 1 ST	Monday 08:30 – 09:20	24/02/2025	AE du Toit Auditorium	ST	Lecture 7	Study Ch 3.2 and do textbook Examples: 3.1, 3.2 Nodal Analysis Do textbook Practice Problems: 3.1, 3.2 Do AMS Assignment # 3 (Deadline: 03 March 2025 at 11:59pm)
	Tuesday 13:30 – 14:20	25/02/2025	AE du Toit Auditorium	ST	Lecture 8	Study Ch 3.3 and do textbook Examples: 3.3, 3.4 Nodal Analysis with Voltage Sources Do textbook Practice Problems: 3.3, 3.4
	Thursday 11:30 – 12:20	27/02/2025	Roos hall	ST	Lecture 9	Study Ch 3.4 and do textbook Examples: 3.5, 3.6 Mesh Analysis Do textbook Practice Problems: 3.5, 3.6
	Friday 07:30 – 08:20	28/02/2025	AE du Toit Auditorium	AL	Tut class #3	Do chapter problems: 3.2, 3.4, 3.12, 3.16, 3.20, 3.30, 3.31
4	Monday 08:30 – 09:20	03/03/2025	AE du Toit Auditorium	AR	Lecture 10	Study Ch 3.5 and do textbook Examples: 3.7 Mesh Analysis with Current Sources Do textbook Practice Problems: 3.7 Do AMS Assignment # 4. (Deadline: 10 March 2025 at 11:59pm)
	Tuesday 13:30 – 14:20	04/03/2025	AE du Toit Auditorium	AR	Lecture 11	Study Ch 3.6-3.7 and do textbook Examples: 3.8, 3.9 Nodal and Mesh Analysis by Inspection Do textbook Practice Problems: 3.8, 3.9
	Thursday 11:30 – 12:20	06/03/2025	Roos hall	AR	Lecture 12	Study Ch 3.9 and do textbook Examples: 3.12, 3.13 Transistors Do textbook Practice Problems: 3.12, 3.13
	Friday 07:30 – 08:20	07/03/2025		AR & AL	Class Test 1	Chapters 1, 2 & 3.2-3.3

Week	Day	Date	Venue	Instructor	Event	Lecture contents and study tasks
5	Monday 08:30 – 09:20	10/03/2025	AE du Toit Auditorium	XY	Lecture 13	Study Ch 4.2-4.3 and do textbook Examples: 4.1, 4.2, 4.3, 4.4, 4.5 <i>Linearity and Superposition</i> Do textbook Practice Problems: 4.1, 4.2, 4.3, 4.4, 4.5 Do AMS Assignment # 5 (Deadline: 17 March 2025 at 11:59pm).
	Tuesday 13:30 – 14:20	11/03/2025	AE du Toit Auditorium	XY	Lecture 14	Study Ch 4.4 and do textbook Examples: 4.6, 4.7 <i>Source Transformation</i> Do textbook Practice Problems: 4.6, 4.7
	Thursday 11:30 – 12:20	13/03/2025	Roos hall	XY	Lecture 15	Study Ch 4.5 and do textbook Examples: 4.8, 4.9, 4.10 <i>Thevenin's Theorem</i> Do textbook Practice Problems: 4.8, 4.9, 4.10
	Friday 07:30 – 08:20	14/03/2025	AE du Toit Auditorium	AL	Tut class #4	Pre-practical Lecture 2 Do chapter problems: 3.37, 3.44, 3.60, 3.69, 3.72, 3.90, 3.91 Do Pre-practical 2 Assignment and submit on AMS (Deadline: 20 March 2025 at 11:59pm).
6	Friday Timetable 07:30 – 08:20	17/03/2025	AE du Toit Auditorium	AL	Tut class #5	Do chapter problems: 4.4, 4.5, 4.11, 4.18, 4.24, 4.37, 4.38, 4.40 Do AMS Assignment # 5 (Deadline: 31 March 2025 at 11:59pm)
	Tuesday 13:30 – 14:20	18/03/2025	AE du Toit Auditorium	XY	Lecture 16	Study Ch 4.6 and do textbook Examples: 4.11, 4.12 <i>Norton's Theorem</i>

						Do textbook Practice Problems: 4.11, 4.12
	Thursday 11:30 – 12:20	20/03/2025	Roos hall	XY	Lecture 17	Study Ch 4.8 and do textbook Examples: 4.13 Maximum Power Transfer Do textbook Practice Problems: 4.13
	Friday	21/03/2025	Public Holiday			
7	Thursday 13:30 – 15:00	27/03/2025	Engineering 3-1 Engineering 3-2 Engineering 3-6 Engineering 3-7	Invigilators	Semester Test 1	Test covers Chapters 1-3. ClickUP announcements will be posted with further details.

Week	Day	Date	Venue	Instructor	Event	Lecture contents and study tasks
8 Practical 2 AR	Monday 08:30 – 09:20	31/03/2025	AE du Toit Auditorium	AR	Lecture 18	Study Ch 5.2-5.3 and do textbook Examples: 5.1, 5.2 Op Amps, Ideal Op Amps Do textbook Practice Problems: 5.1, 5.2 Do AMS Assignment # 6 (Deadline: 07 April 2025 at 11:59pm)
	Tuesday 13:30 – 14:20	01/04/2025	AE du Toit Auditorium	AR	Lecture 19	Study Ch 5.4-5.5 and do textbook Examples: 5.3, 5.4, 5.5 Inverting and Non-inverting Amplifiers Do textbook Practice Problems: 5.3, 5.4, 5.5
	Thursday 11:30 – 12:20	03/04/2025	Roos hall	AR	Lecture 20	Study Ch 5.6-5.7 and do textbook Examples: 5.6, 5.7, 5.8 Summing and Difference Amplifiers Do textbook Practice Problems: 5.6, 5.7, 5.8
	Friday	04/04/2025	AE du Toit	AL	Tut class #6	Do chapter problems 4.48, 4.52, 4.71, 4.72, 5.6, 5.8, 5.9, 5.13, 5.20, 5.31, 5.32, 5.39, 5.45

	07:30 – 08:20		Auditorium			
9	Monday 08:30 – 09:20	07/04/2025	AE du Toit Auditorium	AR	Lecture 21	Study Ch 5.8 and do textbook Examples: 5.9, 5.10 <i>Cascade Circuits</i> Do textbook Practice Problems: 5.9, 5.10 Do AMS Assignment # 7 (Deadline: 22 April 2025 at 11:59pm)
	Tuesday 13:30 – 14:20	08/04/2025	AE du Toit Auditorium	AR	Lecture 22	Study Ch 6.2-6.3 and do textbook Examples: 6.1, 6.2, 6.3, 6.4, 6.5, 6.6, 6.7 <i>Capacitors, Series and Parallel Capacitors</i> Do textbook Practice Problems: 6.1, 6.2, 6.3, 6.4, 6.5, 6.6, 6.7
	Thursday 11:30 – 12:20	10/04/2025	Roos hall	AR	Lecture 23	Study Ch 6.4-6.5 and do textbook Examples: 6.8, 6.9, 6.10, 6.11, 6.12 <i>Inductors, Series and Parallel Inductors</i> Do textbook Practice Problems: 6.8, 6.9, 6.10, 6.11, 6.12
	Friday 07:30 – 08:20	11/04/2025	AE du Toit Auditorium	AL	Tut class #7	Do chapter problems: 5.57, 5.70, 6.10, 6.12, 6.18, 6.40, 6.46, 6.54
14/04/2025 – 18/04/2025 UP Recess						

Week	Day	Date	Venue	Instructor	Event	Lecture contents and study tasks
10	Monday	21/04/2025	Public Holiday			
	Monday Timetable 08:30 – 09:20	22/04/2025	AE du Toit Auditorium	ST	Lecture 24	Study Appendix B and do textbook Examples: B.1, B.2, B.3, B.4, B.5 Complex numbers Do textbook Practice Problems: B.1, B.2, B.3, B.4, B.5 Do AMS Assignment # 8 (Deadline: 29 April 2025 at 11:59pm).
	Thursday 11:30 – 12:20	24/05/2025	Roos hall	ST	Lecture 25	Study Ch 9.2-9.3 and do textbook Examples: 9.1, 9.2 Sinusoids and Phasors Do textbook Practice Problems: 9.1, 9.2
	Friday 07:30 – 08:20	25/05/2025		ST & XY	Class Test 2	Chapters 4, 5, 6 Do Pre-practical 3 Assignment and submit on AMS (Deadline: 02 May 2025 at 11:59pm).
11	Monday	28/04/2025	Public Holiday			
	Tuesday 13:30 – 14:20	29/04/2025	AE du Toit Auditorium	XY	Lecture 26	Study Ch 9.3-9.4 and do textbook Examples: 9.3, 9.4, 9.5, 9.6, 9.7, 9.8 Phasors and Phasor Relationships with Circuit Elements Do textbook Practice Problems: 9.3, 9.4, 9.5, 9.6, 9.7, 9.8
	Monday Timetable 08:30 – 09:20	30/04/2025	AE du Toit Auditorium	XY	Lecture 27	Study Ch 9.5-9.7 and do textbook Examples: 9.9, 9.10, 9.11, 9.12 Impedance, Kirchhoff's Laws in Frequency Domain Do textbook Practice Problems: 9.9, 9.10, 9.11, 9.12
	Thursday	01/05/2025	Public Holiday			
	Friday 07:30 – 08:20	02/05/2025	AE du Toit Auditorium	ST & AL	Tut class # 8	Pre-Practical Lecture 3 Do chapter problems: 9.2, 9.6, 9.13, 9.19, 9.20, 9.42, 9.48, 9.54, 9.66, 9.70

12	Wednesday 15:30 – 17:00	07/05/2025	Engineering 3-1 Engineering 3-2 Engineering 3-6 Engineering 3-7	Invigilators	Semester Test 2	Test covers Chapters 4, 5, 6, 9. ClickUP Announcements will be posted with further details.
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Week	Day	Date	Venue	Instructor	Event	Lecture contents and study tasks
13 Practical 3 ST	Monday 08:30 – 09:20	12/05/2025	AE du Toit Auditorium	XY	Lecture 28	Study Ch 10.2 and do textbook Examples: 10.1, 10.2 Nodal Analysis Do textbook Practice Problems: 10.1, 10.2 Do AMS Assignment # 9 (Deadline: 19 May 2025 at 11:59pm).
	Tuesday 13:30 – 14:20	13/05/2025	AE du Toit Auditorium	XY	Lecture 29	Study Ch 10.3-10.4 and do textbook Examples: 10.3, 10.4, 10.5, 10.6 Mesh Analysis, Superposition Do textbook Practice Problems: 10.3, 10.4, 10.5, 10.6
	Thursday 11:30 – 12:20	15/05/2025	Roos hall	XY	Lecture 30	Study Ch 10.5-10.6 and do textbook Examples: 10.7, 10.8, 10.9, 10.10 Source Transformation, Thevenin & Norton Do textbook Practice Problems: 10.7, 10.8, 10.9, 10.10
	Friday 07:30 – 08:20	16/05/2025	AE du Toit Auditorium	AL	Tut class # 9	Do chapter problems: 10.15, 10.19, 10.36, 10.46, 10.52, 10.56, 10.61, 10.66, 10.68
14	Monday	19/05/2025	AE du Toit Auditorium	ST	Semester Test 3	Optional Semester Test Test covers chapters 6, 9 and 10. ClickUP Announcements will be posted with further details.

	Wednesday	21/05/2025		ST & AL	Aegrotat/Supplementary Pre-Practical	Do Sick/Supp Pre-practical Assignment and submit on AMS (Deadline: 26 May 2025 at 11:59pm). ClickUP Announcements will be posted with further details.
15	Monday - Thursday	26/05/2025 – 29/05/2025	CAEC Labs	ST & AL	Aegrotat/Supplementary Lab Practicals. ClickUP Announcements will be posted with further details.	
31/05/2025 – 19/06/2025 Exam Season						

2 Introduction

This guide serves as Part 2 of the study guide for this module and provides content that is specific to the learning content of the module. Part 1 of the study guide is a General Electrical, Electronic and Computer Engineering (EECE) Study Guide available from the EECE Undergraduate clickUP page or at bit.ly/EECEGeneralUGGuide and provides rules and policies that are applicable to all undergraduate modules presented by the EECE Department. This includes procedures, policies and rules about absence from formal evaluation opportunities and practical sessions, grievance procedures, academic dishonesty and plagiarism.

For EECE students: if you cannot access the current-year EECE Undergraduate Students clickUP module, please arrange access as soon as possible by completing the form at bit.ly/EECEClickUPAccess. All undergraduate-related departmental policies, online resources and communication are managed through this platform.

For students outside the Department of EECE: you do not need access to the EECE Undergraduate Students clickUP as links to relevant documents and online forms are provided in this study guide. Relevant communication via the EECE Undergraduate Students clickUP will be reposted on this module's clickUP Announcements page.

2.1 Welcome

The general objective of this module is to develop expertise in analysing and designing electric and electronic circuits. The module is primarily focused on solving new problems based on known theoretical techniques, as opposed to memorising recipe-type procedures, or solving problems by simply inserting values into formulae. A thorough knowledge of the fundamental theory of circuits will be required. Students will continuously be presented with problems that they might not have seen before, but for which they will have been equipped with the necessary theoretical background. Students are strongly advised to gain as much experience as possible in solving circuit problems from the beginning of the module.

2.2 Educational approach

“For the sole true end of education is simply this: to teach [students] how to learn for themselves; and whatever instruction fails to do this is effort spent in vain.”

Dorothy Sayers

[*The Lost Tools of Learning*](#)

The tools of learning progress as follows: thinking, argumentation, and expression. In the first step a student is shown what the concept/theory/principle is, secondly why and how to use it, and finally the student is allowed to express their knowledge through application to a general problem. More emphasis is placed on *why*-questions rather than *how*-questions. The aim is not only for students to master a specific technology or technique (*how*), but also to become masters of principles and methods so that they may pioneer technologies and techniques (*why*).

A good engineer is one who not only mastered the art of learning, but also enjoys the process of learning new material. Therefore, the lecturers in this module regard their roles as teaching students how to learn for themselves and how to creatively apply their newly acquired skills. If an engineer is to be successful in this world, they should be able to understand the intricacies of a problem, appreciate the underlying assumptions, apply new knowledge in a critical and creative manner, and solve the problem within the logical boundaries of scientific principles.

General approach

The module follows a lecture schedule that will be rigorously adhered to throughout the semester as released on clickUP. The schedule indicates the following:

- the section of the textbook to be discussed during a lecture,
- the set of tutorial problems for each tutorial class,
- the set of textbook problems that must be completed after each class,
- the class test and semester test dates,
- the deadlines for assignments,
- the deadlines for each of the three pre-practical assignments and,
- the week in which practicals will be completed with corresponding pre-practical lectures.

Students are expected to prepare for a lecture by reading through the section of the textbook which will be covered during the lecture. Students should prepare basic notes in preparation for the lecture and attempt the Example Problems. The aim of a lecture is to illustrate and explain the difficult concepts in the textbook. Students can supplement their notes during the lecture. After a lecture, a student is expected to complete the Practice Problems to ensure they understand the concepts associated with a specific textbook section.

Students are expected to complete the tutorial problems in preparation for the tutorial lectures. Since the tutorial lectures will focus on the most difficult problems, not all the problems will be covered during the lectures. However, a complete memorandum will be available after the tutorial class. Students are encouraged to make queries on the Discussion Board on clickUP for problems they struggle with.

The students who do not understand the work after attending lectures, working through Example and Practice Problems in the textbook, and working through the tutorial problems, are welcome to make an appointment with the Lecturers or Assistant Lecturers for consultation. The procedure for consultation is communicated on clickUP. Lectures will, however, not be repeated during the consultation hours. Students must bring their own attempts at solving the problem to the appointment.

There is a set of assignments associated with each chapter in the textbook. These assignments will be made available online and should be completed in the student's own time before the submission deadline as shown in the lecture schedule. Students are encouraged to make queries on the Discussion Board on clickUP for problems they struggle with.

Information regarding the practicals will be communicated on clickUP.

Online approach

The same approach as detailed above will be followed in the case where lectures cannot be presented in-person and on campus except for the following:

- pre-recorded lectures will replace the in-person lecture sessions,
- pre-recorded tutorial videos will replace in-person tutorial sessions,
- online consultation will replace in-person consultation, and
- tests will be written according to a procedure that will be communicated via clickUP.

Hybrid approach

While all lectures, tutorials and practicals will be presented face-to-face in person and on campus, assignments will be completed online. Note that video recordings of lectures and tutorial classes will not be made available. Students are also not allowed to record the in-person lectures or tutorials without the express written consent from the lecturer or assistant lecturer presenting the material.

2.3 Learning in the discipline

Lectures are presented in a style of co-operative and student-centred learning in a face-to-face format on campus by default. Students are expected to prepare for classes. Lecture content and structure will be based on the prescribed and recommended study material. Note, however, that various problems and examples that are not in the textbook will be discussed during lectures. Students are expected to prepare for classes as specified in Section 1.

Attendance of lectures is compulsory and a register of attendance will be taken to determine which students bring their part and work consistently. Refer to the general EECE study guide for the departmental strategy to deal with absence from lectures. Important announcements with respect to laboratory sessions, assignments, tutorials and discussions of test content will be made during lecture times. Students need to take note that only selected components of content, assessments and arrangements delivered in face-to-face lectures will be made available online, e.g., announcements regarding tests, assignments, memoranda (optional), lecture notes (optional), etc. Lecturers are not obliged to publish all content delivered during lectures online as a hybrid teaching and learning strategy will be followed where a significant component of the teaching and learning activities is face-to-face.

As mentioned in Section 2.2, in the case where face-to-face lectures and tutorials are not possible due to specific reasons, pre-recorded videos will be made available via clickUP. Students are expected to follow the same routine by preparing for a lecture, watching a lecture video, and following-up the lecture by completing the necessary textbook problems.

2.4 Restrictions and expectations on the use of generative AI

In this module, the use of generative AI to serve as a tutor support is allowed but not encouraged. In the lecturer's guide: leveraging generative artificial intelligence for teaching and learning enhancement at the University of Pretoria, it says "While generative AI has many strengths, it also has some weaknesses when used in higher education. One of its main limitations is its limited contextual understanding, which can lead to incorrect or irrelevant answers. Its dependence on data quality and

quantity can also lead to biases or limitations, and the lack of references makes it difficult to verify the accuracy of the information provided. Concerns have also been raised regarding the environmental impact, ethical use of training data and bias incorporated into outputs. Therefore, it is essential to carefully evaluate the limitations of generative AI and use it in conjunction with other educational resources, and never to replace practising and learning critical human reflection.”

Should any generative AI being used in EBN 111 assessments, the student needs to complete the form in the appendix of the study guide.

2.5 Responding to student feedback

Students are strongly encouraged to provide feedback on the module via the Student Feedback on Teaching Survey (SFTS) that will be conducted towards the end of the semester. During the semester, feedback may be sent at any time to the lecturer and/or assistant lecturer (either directly or via the class representative) for *immediate* consideration by the lecturers for action.

3 Administrative information

Information that needs to be communicated to students beyond that provided in this guide will be conveyed via announcements made in class, and/or via announcements posted on clickUP.

3.1 Contact details

	Name	Office	Contact details	Consultation
Lecturers	Prof. Xianming Ye	Eng. 1:14-5	012 420 4353 xianming.ye@up.ac.za	Please make an appointment via e-mail
	Mr. Sakhile Twala	Eng. 1: 14-20	sakhile.twala@up.ac.za	
	Mr. Archibald Rohde	Cefim 2-12.8	archie.rohde@tuks.co.za	
Assistant lecturers	Refer to module clickUP page			
EECE undergraduate administration	Ms Bainca Johnson	Eng. 3:7-18	eece.undergraduate@up.ac.za	
Faculty Student Advisor		Your Academic Success Coaches can advise you on goal-setting, adjustment to university life, time management, study methods, stress management and career exploration. Book an individual consultation or attend a workshop. For other support services see section 12 in the EECE General Study Guide.		

Refer to the module clickUP for consultation hours.

3.2 Timetable

The lecture timetable can be found at <https://minstro.up.ac.za/timetables/>, or on your student portal:

Module	Offered	Group	Language	Activity	Day	Time	Venue	Campus	Study Prog
EBN 111	S1	G01	E	L1 L2 L3 L3	Monday Tuesday Thursday Thursday	08:30 - 09:20 13:30 - 14:20 11:30 - 12:20 11:30 - 12:20	AE du Toit Auditorium AE du Toit Auditorium Eng III - 6 Roos hall	HATFIELD	b2c2e2m2n2p2r2s2z2 B1M1R1
EBN 111	S1	G01	E	T1	Friday	07:30 - 08:20	AE du Toit Auditorium	HATFIELD	b2c2e2m2n2p2r2s2z2 B1M1R1

All practicals take place in the CAEC lab (Eng. 2: 4-39) for on-campus practicals. The detailed practical attendance schedules will be posted on clickUP. You will only be in the practical lab three times a semester in accordance with the lecture schedule. The other times that you have a practical slot on your timetable, you will be expected to work on your pre-practical assignments and to build your circuits at home.

Refer to the section on consultation and communication with lecturers in the General EECCE Study Guide.

Kindly refer to the module clickUP page for specific arrangements about consultation hours. Students may consult lecturers and assistant lecturers only during the consulting hours as indicated, or by appointment. Consultation can either be in-person or via a suitable online platform as the situation demands.

This policy also holds before tests and exams. In other words, lecturers and assistant lecturers are only available during their normal consulting hours on the day before a test or examination. This policy aims to encourage students to plan their work and to work continuously.

3.3 Study material and purchases

Prescribed textbooks

Fundamentals of Electric Circuits,
7th Edition
By Charles Alexander and Matthew Sadiku
ISBN10: 1260226409
ISBN13: 9781260226409

Components

Each student requires their own practical kit (cables, components, and breadboard) for the practicals. The cost of the practical kit is included in the class fees. Arrangements will be made for students to receive the kits. Students who repeat the module have the option to not receive new kits and get a reduction from their class fees. Details on applying for this reduction will be administered by the department and queries should be directed towards Ms Bainca Johnson.

4 Module information

4.1 Purpose of the module

The programs presented by the Department of Electrical, Electronic and Computer Engineering have been designed to meet the eleven ECSA graduate attributes (GAs) that qualify the competencies required by professional engineers. While most modules contain at least a small component of a number of ECSA GAs, only the main attributes that are addressed in a module are indicated for the specific module.

This module is presented at entry level for GAs 1 and 2.

Refer to the General EECE Study Guide ([Part 1](#)) for a complete overview of the association of ECSA GAs with the modules in each degree program.

4.1.1 Module description

The general objective of this module is to develop expertise in solving electric and electronic circuits. The topics covered in the module are Ohm's law, Kirchhoff's current and voltage laws, voltage and current division, mesh current and node voltage methods, linearity, Thevenin and Norton equivalent circuits, source transformation, power transfer, energy storage elements in circuits (inductors and capacitors), and operational amplifiers and applications. Although circuits will mostly be solved using direct current (DC) sources, the final part of the module will consider methods to solve circuits using alternating current sources (AC).

4.1.2 ECSA graduate attributes

The critical learning outcomes for the module are qualified through the ECSA graduate attribute (GA) requirements as discussed in the General EECE Study guide. The following skills are developed at the indicated level and a student must demonstrate competence in each to be able to pass the module.

ECSA GA 1 (Entry Level)

Problem solving

Learning outcome: Demonstrate competence to identify, assess, formulate and solve convergent and divergent engineering problems creatively and innovatively.

ECSA GA 2 (Entry Level)

Application of scientific and engineering knowledge

Learning outcome: Demonstrate competence to apply knowledge of mathematics, basic science and engineering sciences from first principles to solve engineering problems.

4.1.3 UP graduate attributes

The following UP graduate attributes will be nurtured in this module: capability for autonomous work, technical communication abilities, and the practical skills component of knowledge acquisition.

Refer to the section on cognitive level of assessment in the General EECE Study Guide.

	%*
--	----

1.	Knowledge	10
2.	Comprehension	10
3.	Application	60
4.	Analysis	10

* Estimate of the % of the total assessment, including all forms of assessment applied in this module, devoted to the various levels of cognitive thinking skills and of UP graduate attributes.

** Assessment of UP graduate attributes:

- Practical skills
- Independent working skills
- Time management
- Conceptualize and synthesize knowledge

4.2 Articulation with other modules in the programme

Detailed diagrams that show the inter-relationship and logical sequence of the various modules in the study programme and the EBN 111 module within the context of the programme, are available on the current year EECE Undergraduate Student clickUP page. Under “Course Content”, go to “Academic matters” > “Degree Programs and Modules” > “Curricula”.

4.3 Learning presumed to be in place

Students who are admitted to the relevant engineering programmes are encouraged to take this module.

4.4 Overall competencies/module outcomes

Apart from the aforementioned ECSA and UP graduate attributes, at the completion of the module, students should know, understand and be able to apply basic circuit analysis tools to both DC and AC circuits. The analysis tools include: Ohm’s Law, Kirchhoff’s Laws, Nodal and Mesh Analysis, Superposition, Source Transformation, and Thevenin and Norton equivalents. At the completion of the module, students should be able to solve any DC or AC circuit containing resistors, capacitors, inductors, sources and operational amplifiers using the circuit analysis tools mastered during the module. Assessments during the module are purely problem-based, e.g., students need to solve for either voltages or currents in circuits.

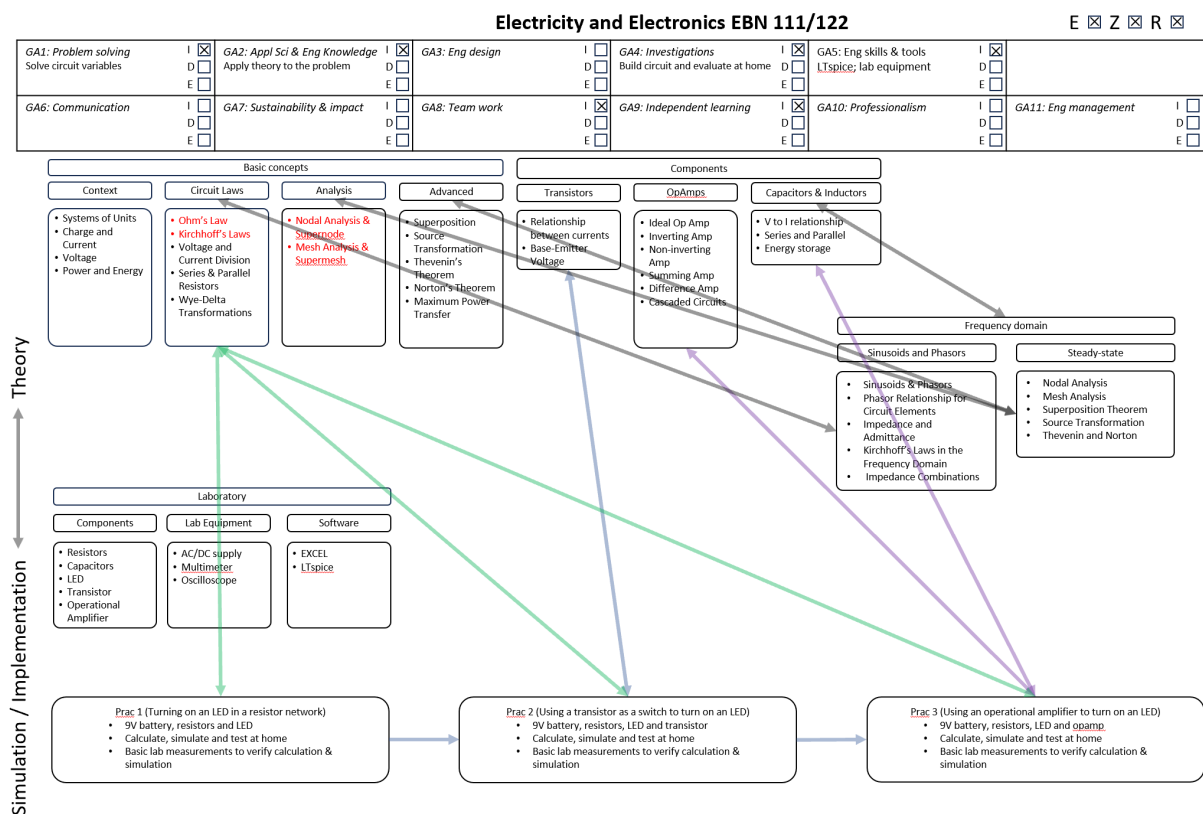
4.5 Module structure

Study theme and study units	Mode of instruction	Chapter
1. Basic Concepts • Systems of Units • Charge and Current • Voltage • Power and Energy • Circuit Elements • Applications	Lectures, class discussion, tutor classes and self-study.	1

Study theme and study units	Mode of instruction	Chapter
2. Basic Laws • Ohm's Law • Nodes, Branches, and Loops • Kirchhoff's Laws • Series Resistors and Voltage Division • Parallel Resistors and Current Division • Wye-Delta Transformations • Applications	Lectures, class discussion, tutor classes and self-study.	2
3. Methods of Analysis • Nodal Analysis • Nodal Analysis with Voltage Sources • Mesh Analysis • Mesh Analysis with Current Sources • Nodal and Mesh Analysis by Inspection • Nodal Versus Mesh Analysis • Transistors • Applications	Lectures, class discussion, tutor classes and self-study.	3
4. Circuit Theorems • Linearity Property • Superposition • Source Transformation • Thevenin's Theorem • Norton's Theorem • Derivations of Thevenin's and Norton's Theorem • Maximum Power Transfer • Applications	Lectures, class discussion, tutor classes and self-study.	4
5. Operational Amplifiers • Operational Amplifiers • Ideal Op Amp • Inverting Amplifier • Non-inverting Amplifier • Summing Amplifier • Difference Amplifier • Cascaded Op Amp Circuit • Applications	Lectures, class discussion, tutor classes and self-study.	5
6. Capacitors and Inductors • Capacitors • Series and Parallel Capacitors • Inductors • Series and Parallel Inductors • Applications	Lectures, class discussion, tutor classes and self-study.	6

Study theme and study units	Mode of instruction	Chapter
7. Sinusoids and Phasors • Sinusoids • Phasors • Phasor Relationship for Circuit Elements • Impedance and Admittance • Kirchhoff's Laws in the Frequency Domain • Impedance Combinations • Applications	Lectures, class discussion, tutor classes and self-study.	9
8. Sinusoidal Steady-State Analysis • Nodal Analysis • Mesh Analysis • Superposition Theorem • Source Transformation • Thevenin and Norton Equivalent Circuits • Applications	Lectures, class discussion, tutor classes and self-study.	10
9. Laboratory work	Computer-assisted and experimental work in laboratories.	

The module structure can be summarized by means of the following module map:



4.6 Credit map and notional hours

The number of credits allocated to a module gives an indication of the volume of learning required for the completion of that module and is based on the concept of notional hours. Should a module carry a weighting of 16 credits, it follows that you should spend an average of 16×10 hours of study in total on the module (1 credit = 10 notional hours). This includes time for lectures, assignments, projects, tests and exams. This means that you should spend approximately $160 \text{ hours} / 14 \text{ weeks} = 11.5 \text{ hours per week}$.

Lectures, tutorials and practicals (contact time)					
	Contact time per week (periods)	Preparation per week (periods)	Total time per week (hours)	Number of weeks	Total time per semester (hours)
Lectures*	3	3	5	12	60
Tutorials*	1	3	3.3	12	40
Practicals	1	1	1.5	12	18
Assignments and assessment (non-contact time)					
	Assessment time (hours)	Preparation time (hours)			Total time per semester (hours)
Assignment(s)	0	8			8
Semester test 1	1.5	7.5			9
Semester test 2	1.5	7.5			9
Examination	3	15			18
Module hours					162
Module credits					16.2
Note 1: The time allocation in this table represents nominal hours, i.e., some students might require less and others more time to attain the pass requirements.					
* Note 2: As class tests are written during lecture or tutorial periods, the times allocated for class tests are included in the total lecture or tutorial time.					

4.7 Study units (study themes)

Refer to the module calendar for dates of the contact sessions during which the units will be discussed.

STUDY THEME 1: Basic Concepts

Learning outcomes

The student will:

- Understand the basic SI units and SI prefixes
- Learn about the concepts of charge, current, and voltage
- Learn about power and energy
- Become familiar with the passive sign convention
- Learn about the circuit elements including dependent and independent sources

Study units

Chapter 1, section 1.1: Introduction

Chapter 1, section 1.2: SI units

Chapter 1, section 1.3: Charge and Current

Chapter 1, section 1.4: Voltage

Chapter 1, section 1.5: Power and Energy

Chapter 1, section 1.6: Circuit Elements

Chapter 1, section 1.7 and section 1.8: Applications and Problem Solving (self study)

Criteria of assessment

Upon completion of this study theme, the student will:

- **know** the SI units and be able to **determine** derived units.
- know the different equations and units for voltage and current and be able to apply them.
- be able to **define** the basic circuit element.
- **know** the passive sign convention and be able to **apply** it.
- **know** the equation for power and be able to **apply** it.
- be able to **derive** the equation for power in terms of the basic circuit element's terminal voltage and current and be able to **apply** it.
- **know** the characteristics of ideal voltage and current sources.
- be able to **apply** and **interpret** the passive sign convention in power calculations.
- be able to **compute** charge, voltage, current, power and energy for an element.

STUDY THEME 2: Basic laws

Learning outcomes

The student will:

- Learn about Ohm's law
- Understand the concepts of nodes, branches, and loops
- Learn Kirchhoff's current and voltage laws
- Learn about combining resistors in series and in parallel
- Learn about the principles of voltage division and current division
- Become familiar *with wye-delta transformation*

Study units

Chapter 2, section 2.2: Ohm's Law

Chapter 2, section 2.3: Nodes, Branches, and Loops

Chapter 2, section 2.4: Kirchhoff's Law

Chapter 2, section 2.5: Series Resistors and Voltage Division

Chapter 2, section 2.6: Parallel Resistors and Current Division

Chapter 2, section 2.7: Wye-Delta Transformations

Chapter 2, section 2.8: Applications (self study)

Criteria of assessment

Upon completion of this study theme, the student will:

- **know** the five ideal circuit elements.

- be able to **apply** the definitions of ideal sources and hence **determine** the validity of connected sources.
- **understand** the way dependent sources are handled.
- **understand** the difference between passive and active elements.
- **know** Ohm's law and be able to **apply** it.
- **know** the equations for power in terms of resistance and conductivity and be able to **apply** them.
- be able to **construct** circuit models and **identify** shortcomings with respect to models of practical systems.
- be able to **identify** nodes, loops and meshes.
- be able to **define** direct current, series, parallel and other basic concepts.
- be able to **derive** equations for elements equivalent to elements connected in series and/or parallel.
- be able to **solve** problems involving voltage and current division.
- be able to **design** voltmeters and ammeters.
- **know** Kirchhoff's voltage and current laws and be able to **apply** them.
- be able to **derive** equations for delta-to-Y conversions and **apply** them

STUDY THEME 3: Methods of Analysis

Learning outcomes

The student will:

- Learn about nodal and mesh analysis
- Understand the concept of supernode and supermesh
- Apply what is learnt to DC transistor circuits

Study units

Chapter 3, section 3.2: Nodal Analysis

Chapter 3, section 3.3: Nodal Analysis with Voltage Sources

Chapter 3, section 3.4: Mesh Analysis

Chapter 3, section 3.5: Mesh Analysis with Current Sources

Chapter 3, section 3.6: Nodal and Mesh Analysis by Inspection

Chapter 3, section 3.7: Nodal Versus Mesh Analysis

Chapter 3, section 3.9: Applications: DC Transistor Circuits (self study)

Criteria of assessment

Upon completion of this study theme, the student will be able to:

- determine a set of independent simultaneous equations consisting of node-voltage equations and constraint equations from which a circuit can be solved.
- **identify** a supernode and **formulate** a new set of equations associated with the supernode.
- **determine a set of** independent simultaneous equations consisting of mesh-current equations and constraint equations from which a circuit can be solved.
- **identify** a supermesh and **formulate** a new set of equations associated with the supermesh.
- **identify** the constraint equations and **formulate** the new set of equations associated with dependant sources for the mesh-current method.

STUDY THEME 4: Circuit Theorems

Learning outcomes

The student will:

- Learn about theorems used for simplifying analysis
- Understand the principles of linearity and superposition
- Learn of about the concepts of source transformation and equivalence.
- Be able to use Thevenin's theorem to find a Thevenin equivalent circuit.
- Be able to use Norton's theorem to find a Norton equivalent circuit.
- Understand the concept of maximum power transfer to the load.

Study units

Chapter 4, section 4.2: Linearity Property

Chapter 4, section 4.3: Superposition

Chapter 4, section 4.4: Source Transformation

Chapter 4, section 4.5: Thevenin's Theorem

Chapter 4, section 4.6: Norton's Theorem

Chapter 4, section 4.7: Derivations of Thevenin's and Norton's Theorem

Chapter 4, section 4.8: Maximum Power Transfer

Chapter 4, section 4.10: Applications (self study)

Criteria of assessment

Upon completion of this study theme, the student will be able to:

- **use** the linearity property to **calculate** circuit variables
- **use** the superposition principle to **calculate** circuit variables
- **derive** a source transformation for a given circuit and **use** source transformation to **calculate** circuit variables.
- **calculate** Thévenin and Norton equivalent circuits by **determining** which of the following methods is applicable and then **apply** it:
 - **determine** V_{Th} and R_{Th} by **calculating** the open-circuit voltage and short-circuit current.
 - **apply** source transformations
 - **determine** R_{Th} by the method of deactivating independent sources
 - **calculate** R_{Th} by deactivating independent sources and adding a test source for circuits containing dependant sources only.
- **use** the superposition principle to **derive** Thevenin's and Norton's Theorems
- **derive** the equations associated with maximum power transfer and **apply** the underlying principles to other circuits.
- **apply** the method of superposition to linear circuits.

STUDY THEME 5: Operational amplifiers.

Learning outcomes

The student will:

- Understand what operational amplifiers are
- Learn about the ideal op amp
- Learn some op amp circuits including inverting, noninverting, summing, and difference amplifiers

Study units

Chapter 5, section 5.2 and section 5.3: Operational Amplifier and Ideal Op Amp

Chapter 5, section 5.4: Inverting Amplifier

Chapter 5, section 5.5: Non-Inverting Amplifier

Chapter 5, section 5.6: Summing Amplifier

Chapter 5, section 5.7: Difference Amplifier

Chapter 5, section 5.8: Cascaded Op Amp Circuit

Chapter 5, section 5.10: Applications (self study)

Criteria of assessment

Upon completion of this study theme, the student will be able to

- **derive** transfer characteristics of basic op-amp circuits (e.g., inverting-amplifier circuit, summing-amplifier circuit, noninverting-amplifier circuit, and difference-amplifier circuit) from first principles.
- completely **solve** circuits in which no more than four op-amps are connected.

STUDY THEME 6: Capacitors and Inductors

Learning outcomes

The student will:

- Learn about capacitors and capacitance
- Understand how to combine capacitors in series and parallel
- Learn about inductors and inductance
- Understand how to combine inductors in series and parallel

Study units

Chapter 6, section 6.2: Capacitors

Chapter 6, section 6.3: Series and Parallel Capacitors

Chapter 6, section 6.4: Inductors

Chapter 6, section 6.5: Series and Parallel Inductors

Chapter 5, section 6.6: Applications (self study)

11.6.3. Criteria of assessment

Upon completion of this study theme, the student will be able to

- **solve** time-varying voltages and currents in circuits containing combinations of either inductors or capacitors.
- **Compute** energy and power associated with the inductors and capacitors in the above problems.

STUDY THEME 7: Sinusoids and Phasors

Learning outcomes

The student will:

- Understand the concepts of sinusoids and phasors
- Apply phasors to circuit elements
- Be introduced to the concepts of impedance and admittance
- Learn about impedance combinations

Study units

Chapter 9, section 9.2: Sinusoids

Chapter 9, section 9.3: Phasors

Chapter 9, section 9.4: Phasor Relationship to Circuit Elements

Chapter 9, section 9.5: Impedance and Admittance

Chapter 9, section 9.6: Kirchhoff's Laws in the Frequency Domain

Chapter 9, section 9.7: Impedance Combinations

Chapter 9, section 9.8: Applications (self study)

Criteria of assessment

Upon completion of this study theme, the student will be able to:

- **compute** the phasor representation of sinusoidal currents and voltages and also determine inverse phasor transforms.
- **compute** the rms-value of a periodic function.
- **Apply** Kirchhoff's Law in the frequency domain

STUDY THEME 8: Sinusoidal Steady-State Analysis

Learning outcomes

The student will:

- Apply previously learnt circuit techniques to sinusoidal steady-state analysis
- Learn how to apply nodal and mesh analysis in the frequency domain
- Learn how to apply superposition, Thevenin's and Norton's theorems in the frequency domain

Study units

Chapter 10, section 10.2: Nodal Analysis

Chapter 10, section 10.3: Mesh Analysis

Chapter 10, section 10.4: Superposition Theorem

Chapter 10, section 10.5: Source Transformation

Chapter 10, section 10.6: Thevenin and Norton Equivalent Circuits

Chapter 10, section 10.7: Applications (self study)

Criteria of assessment

Upon completion of this study theme, the student will be able to:

- **compute** the frequency domain equivalent circuit of any circuit containing sinusoidal sources and combinations of resistors, inductors, and capacitors.
- **solve** frequency domain circuits by using standard circuit techniques.

STUDY THEME 9: Practicals

Learning outcomes

The student will:

- Analyse a set of circuits using previously learnt circuit theorems
- Simulate the circuits with an appropriate computer assisted design tool.
- Build the circuits on an experimental board and evaluate the circuits in a laboratory setup.

Criteria of assessment

Upon completion of this study theme, the student will be able to:

- **use** computer aided software tools to analyse a circuit.
- **use** basic laboratory tools to evaluate a physical circuit.

4.8 Assessment

4.8.1 Requirements, subminima and calculation of marks

Primary Requirements	1. The examination schedule per module is compiled in accordance with the Examination Regulations of the Faculty, the General Examination regulations of the University and the Policy on Examinations and related matters. 2. Admission to examination: To be admitted to the final examination in a module, a semester/year mark of at least 40% is required, with the exception of first year modules on the first semester level, where a semester mark of at least 30% is required. Completion of all assessments that count towards the semester mark is mandatory for entry into the examination of a module. In addition, all other examination admission requirements must be complied with, as applicable to the specific module. 3. Minimum examination mark: In order to pass a module a student must obtain an examination mark of at least 40% and a final mark of at least 50%. 4. ECSA graduate attribute requirements: In modules where ECSA GAs are assessed at exit level, the relevant GA(s) must be passed in addition to the general pass requirements stated above to pass the module. GA pass requirements are stated in the details of the assessment schedule.
Calculation of final mark	Semester mark: 50% Examination mark: 50%
Calculation of semester (year) mark I = individual; G = group	Semester test 1: 35% (I) Semester test 2: 35% (I) Assignments: 6% (G) Class tests: 12% (I) Practicals: 12% (G)
Specific requirements	All scheduled practical assignments must be completed to a satisfactory level. All semester projects and tasks must be completed to a satisfactory level as per Departmental rules. Failure to comply with any of the above will result in examination refusal. Additional conditions: 1. Semester tests 1 and 2 are compulsory. 2. A voluntary third semester test (Semester Test 3) may be written at the end of the semester. The highest two marks of the three written semester tests will be used to calculate the semester mark. No sick test will be awarded for the optional Semester Test 3. Students that have qualified to write a sick test for Semester Tests 1 and/or 2, may be required to write a sick test at the same time as Semester Test 3. As such, students who qualified to write a sick test will not be eligible to write Semester Test 3.

Subminimum requirements	As per primary requirements above. Additional subminimum requirement (semester): 1. An average practical mark of at least 50% must be attained. 2. A subminimum of 40% must be attained for each practical assignment. Based on the qualification criteria in the practical guide, one additional demonstration opportunity will be provided for each practical assignment for students who participated in the practical and did not comply with the 40% practical subminimum for the practical. The average practical mark of students who make use of the additional opportunity will be capped at 50%. Additional subminimum requirements (examination/final mark): None
Requirements for promotion	No promotion allowed.

4.8.2 Assessment details

Assessment approach

Assessment of class tests, semester tests and the exam will take place electronically via the AMS (Assessment Management System). See module clickUP page for details.

4.8.2.1 Class tests

Two (short) class tests will be written in the semester. The dates for the tests will be announced in class and on the clickUP page. A mark of zero will be awarded if a student misses a class test, unless the student can provide a valid medical certificate. For legitimate absence the test does not count towards the semester mark. Illegitimate absence from a class test does not result in exam refusal, but the test counts towards the semester mark.

4.8.2.2 Semester tests

Two compulsory tests of 90 minutes each will be written during the scheduled test weeks of the School of Engineering.

For a legitimate absence from these tests as supported by the necessary documentation (see [section 8.5](#) in the EECE General Study Guide), an aegrotat test will be written in the final week of lectures.

A voluntary third test of 90 minutes will be written. The mark for this test will replace the lowest mark of the first two tests. There is no aegrotat test for the voluntary third test.

The AMS (an online marking system developed by the EECE department) will be applied to mark the class tests, semester tests and exams. The online marking system is accessible via: <https://ams.up.ac.za/>. Each student will be granted an account to access the online marking system, where students can view their marks, marking log, memos of each question, and make queries for any marking discrepancies. A certain marking query period will be allocated after each class test, semester test, and exam. **Students must please ensure that they visit the AMS online marking system during the granted query period.** After the query period, the mark that appears on the module web page for the relevant test will be considered as final. **No late queries on tests/exams are allowed.**

4.8.2.3 Optional test

Refer to [section 8.5.5](#) in the EECE General Study Guide for the alternative approach and rules that will be followed for sick tests this semester.

4.8.2.4 Examination

Tests and examinations are **CLOSED** book for on-campus evaluations.

4.8.2.5 Assignments

A series of online assignments will be released during the semester. The assignments will be released on the Assessment Management System (AMS) (<https://ams.up.ac.za/users/login/>). Details on how to access and complete assignments on the AMS will be provided on clickUP and in the EECE General Study Guide. Students need to complete the assignments online before their respective deadlines as published on the AMS. Refer to [section 8.4](#) of the EECE General Study Guide for the online submission protocol related to deadlines. Failure to complete an assignment will result in a mark of zero for the assignment.

The assignments contribute to the semester mark and the submission protocol is provided in [section 8.4.2](#) of the EECE General Study Guide. Failure to submit an assignment will result in a mark of zero for the particular assignment. There will not be any aegrotat assignment opportunities. Students who are repeating the module are expected to redo all the assignments.

If there is suspicion that any part(s) of the work has been copied from another student's work or from another source, the relevant UP policies on plagiarism and academic dishonesty will be followed. Refer to Section 9.2 in the General EECE Study Guide for more information about academic misconduct and the processes that are required to deal with such incidents.

You will be required to sign the following declaration on the front page of your assignments:

I, name and surname, hereby declare that the work herewith is completely my own and that no parts have been copied in any way from current or previous students or other sources.

The use of a chatbot such as ChatGPT is allowed, but needs to be acknowledged explicitly.

4.8.2.6 Practicals

A practical is the combination of a pre-practical assignment and a laboratory assignment. Each assignment is submitted and evaluated via AMS.

Three three-hour practicals will be carried out: for dates, venues, practical guides and content descriptions, and submission requirements, refer to *Practicals* on the module clickUP page.

The three practicals will all be laboratory session practicals (i.e. experimental model practicals; please see the General EECE Study Guide Section 7.1 for a definition). This means that the practical time is divided into three 3-hour laboratory sessions during which students conduct experiments or computer simulations, or develop computer algorithms.

Practicals 2 and 3 will be done individually by each student.

Please consult the EECE Undergraduate clickUP website for general procedures regarding the practical sessions, such as the laboratory rules and after-hours access to the labs.

It is not allowable for a student to use their practical work from previous years if the module is repeated. In this module, repeaters are allowed to apply for practical exemption if they previously passed the practical component section, but did not pass the overall module. These students' final practical mark will be transferred from their most recent failed attempt of the module. Details on the application for practical exemptions will be communicated on clickUp.

The practical sessions are compulsory: failure to attend a practical session will deny you permission to write the final examination.

Approach: *Model 1: Laboratory session practicals*

As per the General EECE Study Guide, the module will make use of Model 1 for practicals. In other words, the practical time is divided into three 3-hour laboratory sessions during which students conduct experiments and complete a laboratory assignment.

A practical is considered as the combination of the pre-practical assignment and the laboratory assignment. Each laboratory session is preceded by a compulsory pre-practical assignment. As indicated in Section 6.1, three hours of preparation time is necessary to complete a pre-practical assignment. Each student is expected to present a functioning circuit in order to gain entry into the CALC lab for each practical. The details on which circuit to build at home will be found in the pre-practical assignment. All the lab rules for the EECE Department must be followed, including but not limited to, wearing closed shoes and not flip flops or sandals, not wearing caps and not bringing in any food or drinks into the lab, including water bottles.

Practicals:

1. Practical 1: Turning on an LED in a resistor network
2. Practical 2: Using a transistor as a switch to turn on an LED
3. Practical 3: Using an operational amplifier to turn on an LED

The practical guides are made available on clickUP. The pre-practical and laboratory assignments are completed and evaluated on the AMS. Note that students are encouraged to work in groups of two during laboratory session, but will complete the pre-practical and laboratory assignments on the AMS on an **individual** basis.

Students repeating the module: If a student is repeating the module and has not been granted practical exemption, the student may not use the practical work or reports from the previous year.

Rules and requirements: Each student must do three practicals (pre-practical & laboratory assignment). In order to pass this module,

1. all practicals must be done,
2. an average mark of 50% or more must be obtained overall for the practicals, and
3. a subminimum of 40% must be attained for each practical.

Laboratory groups and sessions: Students will be divided into large laboratory session groups by engineering study discipline (e.g., electronic, mechanical) at the start of the semester. The dates and times each group has to do a particular laboratory session will appear on clickUP. The responsibility rests on each student to make sure when laboratory session are scheduled for them.

Permission to be moved from a laboratory session group must be obtained in writing from the Assistant Lecturers at least one week before the date of the laboratory session. Failure to pre-arrange

a move from one group to another will mean that the student will not be able to do the laboratory session and will subsequently not be eligible for entry into the examination in the module.

A student needs to do all practicals and to be admitted to the exam. In the event of absence from a practical, please refer to the protocol in Section 8.5 of the EECE General Study Guide.

During laboratory sessions: Students must be well prepared when reporting for a laboratory session. The information on clickUP as well as the relevant material in the textbook must be consulted beforehand. Each student must bring their own textbook to the practical. Students must obey all laboratory rules.

Information regarding practicals: Specific information and about practicals will be posted on clickUP at least two weeks prior to the commencement of the particular practical. General announcements and reminders will also be made on clickUP.

Use of chatbots: The use of a chatbot such as ChatGPT is allowed, but needs to be acknowledged explicitly.

Practical fallback strategy for single students

The practical sessions in this module are completed individually. As such a single student fallback strategy is not applicable.

Contingency measure for students unable to attend on-campus practicals owing to serious unforeseen circumstances

Students who are unable to participate in on-campus practicals in the event that such practicals can be presented, must contact the lecturer to discuss a contingency arrangement.

Approach to online practicals

The approach to practicals described above assumes practicals can be completed on campus. If the situation warrants, practicals will be completed online. Any amendments to the submission and completion of practicals will be communicated on clickUP.

Appendix A: GEN AI USAGE FORM FOR ASSIGNMENTS

Q1. AI Declaration [tick one box]

1. I used Generative Artificial Intelligence in the current assignment	
2. I did not use any Gen AI for the current assignment	

Ensure you are permitted to use generative AI by reading the instructions for your assignment or assessment or by consulting your study guide. Failure to follow the instructions regarding using generative AI for your assignment constitutes academic dishonesty. Examples of generative AI include ChatGPT, Google's Gemini, Microsoft's Copilot, Claude, and Meta's Llama 3 AI integrated into WhatsApp and Facebook. If you selected option 1 in Q1, please continue to complete the rest of the form:

Q2. Student Information

Name & Surname	
Student Number	
Course Code	
Assignment Title	

Q3. Prompts used

Please paste all the prompts you created for the assignment and indicate which aspect(s) of the assignment the specific prompt was used for.

Prompt	Aspect of Assignment

Q4. Type of usage

Briefly describe the features for which you used the Gen AI, you may say "Not Applicable" where needed:

Feature used	Description
Brainstorming and idea generation	
Language editing suggestions	
Feedback and revision suggestions	
Explaining complex concepts	
Writing coach	
Other (please specify):	

Q5. Ethical use

Write a brief paragraph explaining how you ensured the usage of Gen AI was aligned with the ethical and responsibility requirements of the University of Pretoria ([link](#)). Consider examples such as repurposing and reintegrating ideas generated by Gen AI with your own thoughts, integrating Gen AI ideas with other literature, critically evaluating Gen AI outputs, maintaining transparency about Gen AI usage, enhancing your learning and ensuring comprehension despite using Gen AI, and personal development through using Gen AI as an assistant. If you directly used text or data generated by Gen AI, ensure it was cited appropriately.

Ethical use statement:

Q6. Why did you use Gen AI?

Signature:

Student Signature	
Date	